Improved ANN-Based Algorithm for Detection and Classification of Faults on Transmission Lines

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Abstract*— This paper presents a relaying algorithm based on Artificial Neural Network (ANN) technique for the protection of transmission line. A feed forward ANN with six inputs and eleven outputs has been developed for the detection and classification of faults. Data was generated by simulating a 400 kV, 50Hz, 100 km transmission line in PSCAD/EMTDC at a sampling frequency of 2 kHz. Three ANN configurations with different combinations of inputs have been attempted. Initially all the three ANN configurations were trained and tested using truncated data for their comparative performance. ANN-2 configuration has been found to be the best one and has an accuracy of 100% for fault detection and classification in both training and testing phases with the relay operating time of 12.5 ms. ANN-2 has been further trained and tested using full data. Two-fold cross verification was carried out. An accuracy of 100% was obtained on testing with 12.5 ms delay each time.

***Keywords*— artificial neural network, fault detection, phase selection, power system faults, transmission line protection.**

I. Introduction

A transmission line is one of the main components in electric power system which provides a path to transfer power from generation to load. The transmission line is exposed to different types of faults namely phase-to-earth, two-phase-to-earth, phase-to-phase, three phase faults. The fault on the transmission line needs to be detected and then isolated quickly to prevent the disturbance in the line. It is important to detect, classify and clear the faults with high speed, selectivity and accuracy in order to prevent instability which can cause damage to the system. Several techniques have been reported in the literature for detection and location of faults on transmission lines [1,2]. The objective is to protect the transmission line in such a way that the power system is stable and only the faulty part is to be isolated leaving the rest part of the system in operation. Hence, it is essential to detect the fault quickly and separate the faulty section of the transmission line for safety, economy and power quality.

A fault detector that uses artificial neural networks (ANN) has been described in [3]. It performs the fault defection problem as a pattern classification process. However it does not classify the faults. Back-propagation ANN has been used as a pattern classifier for a distance of 100 km with two sources at both ends in [4]. However, only one type of fault, phase to ground fault, was simulated varying the fault distance, fault resistance, fault inception angles, sources impedance, source capacities and power transfer angle. ANN used six inputs, namely normalized I_a, I_b, I_c and V_a, V_b, V_c and had six nodes in the input layer, six nodes in the first hidden layer, two nodes in the second hidden layer and one node in the output layer.ANN was successfully trained to detect fault in protection zone only with accuracy at 78.82% of the protection zone. Fault detection and classification has been achieved in [5]. Two hidden layers with (30-20-15-11) have been used. However, it requires large memory as the number of inputs is more.

Faults have been detected and classified using ANN with the relay operating time of 1 cycle in [6]. Single-line-ground faults have been classified with the relay operating time of 1 cycle in [7].

The prime motive behind the present work was an accurate fault classification as well as fault detection in such a way that the maximum accuracy is achieved with minimum relay operating time. Paper reports improved algorithm which does classification as well as detection and has accuracy of 100% and requires less memory with relay operating time of less than ¾ cycle which is an improvement over work achieved by the earlier researchers.

ANN has been trained and tested using Back-propagation algorithm with a large amount of data. The inputs to ANN were the rms values of phase voltages, line voltages, line currents and ratios of sequence components of currents in different combinations.

The transmission line was simulated using an electromagnetic transient program, EMTDC/PSCAD on a sample three-phase power system. The data has been generated for all the types of faults at two different locations with two values of fault resistance and eight values of fault inception angles at a sampling rate of 2 kHz.
Three ANN configurations (ANN-1, ANN-2, ANN-3) with six inputs and eleven outputs have been attempted. The three configurations differ in the inputs. The inputs were V_a, V_b, V_c and I_a, I_b, I_c to ANN-1, V_{ab}, V_{bc}, V_{ca} and I_n/I_p, I_o/I_p, I_p/I_{load} to ANN-2 and. Va, Vb, Vc and I_n/I_p, I_o/I_p, I_p/I_{load} to ANN-3.

The eleven outputs include AG, BG, CG, AB, BC, CA, ABG, BCG, CAG, ABCG and no-fault. One hidden layer was used with 20 neurons.

All the three ANN configurations were attempted and compared. ANN-2 has been found as the best configuration in terms of accuracy and performance as 100% accuracy has

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been achieved in both training and testing with a minimum delay of 12.5 ms.

II. Data Generation For Training and Testing of ANNs

A. Simulation

Simulations were performed using an electromagnetic transient program, EMTDC/PSCAD, on a sample three-phase power system. Fig. 1 shows the three phase model of the system that has been studied. The system consists of a generator of 400 kV located on one side of the transmission line. The transmission line (line-1 and line-2 together) is 100 km long. The line has been modeled using distributed parameters. A three phase fault module is used to simulate different types of faults with different inception angles, different fault resistances, and different source and load impedances at various locations on the transmission line.

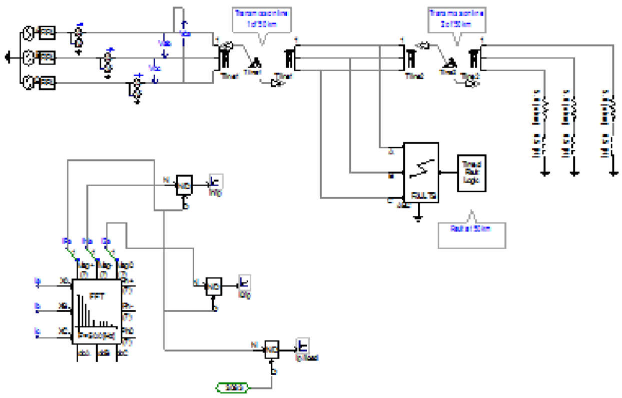

Fig. 1. Model simulated in PSCAD (fault location at 50km)

B. Data Needed

As the input signals to the protective relay are currents and voltages. In the same way these inputs have to be fed to the ANN. However, some other parameters are needed to differentiate between the types of faults. Those parameters are zero sequence currents (I_o), positive sequence currents (I_p) and negative sequence currents (I_n). Twelve quantities are needed to select the best set of input combinations for the neural network. The quantities include I_a, I_b, I_c, V_a, V_b, V_c, V_{ab}, V_{bc}, V_{ca} and I_n/I_p, I_o/I_p, I_p/I_{load}. Balanced faults and unbalanced faults are differentiated by the ratio of negative sequence current to positive sequence currents (I_n/I_p), the ratio of zero sequence currents to positive sequence currents (I_o/I_p) is used to differentiate between the ground faults to phase faults and the balanced fault and no fault condition is differentiated by the ratio of positive sequence currents to load currents (I_p/I_{load}).

C. Generated Data

The values of three-phase rms voltages and currents are measured. The rms voltages include the rms line to line voltages and rms phase voltages. The line components are first processed using FFT algorithm and then the sequence components at the fundamental frequency are derived. Fig. 2 shows the graphs obtained during single line-to-ground fault of the system.

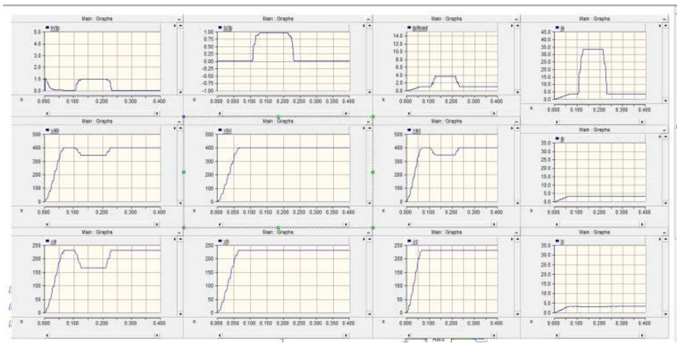

Fig. 2. Rms graphs of 12 parameters during single-line- to-ground fault

III. ANN Inputs and Configuration

Three ANN configurations have been attempted. The configurations differ in the input signals applied. The three combination of inputs include (V_a, V_b, V_c and I_a, I_b, I_c) for ANN-1, (V_{ab}, V_{bc}, V_{ca} and I_n/I_p, I_o/I_p, I_p/I_{load}) for ANN-2 and (V_a, V_b, V_c and I_n/I_p, I_o/I_p, I_p/I_{load}) for ANN-3.

A. ANN-1

1) Inputs

Phase Voltages and Line Currents were used as Inputs to ANN-1. The six inputs used include V_a, V_b, V_c and I_a, I_b, I_c. The outputs were eleven which consists of AG, BG, CG, AB, BC, CA, ABG, BCG, CAG, ABCG and no-fault. One hidden layer was used with different number of neurons for optimization.

2) ANN-1 Configuration

Fig. 3 shows the configuration of ANN-1 with six inputs, one hidden layer of 20 neurons and 11 outputs.

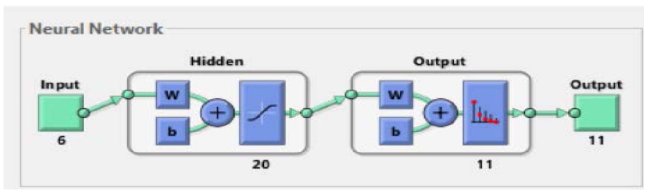

Fig. 3. ANN-1 (6-20-11) Configuration

3) Optimization of number of neurons in hidden layer

In ANN-1, the network takes in six inputs at a time, which are the rms phase voltages and line currents for all the three phases (scaled with respect to the pre-fault values) for ten different faults and also no-fault case. The number of neurons in the hidden layer was varied. First it was set at 10 neurons. Then the number of neurons in the hidden layer was increased for optimization from 10 neurons to 15neurons and from 15 neurons to 20 neurons and then to 25neurons. Research was carried out and the optimized number of neuron in the hidden layer has been found to be 20.

4) Training and Testing of ANN-1

The pattern recognition and classification abilities of neural networks has been utilized to develop ANN based relay algorithm for the protection of transmission line. ANN-1 consists of six inputs namely V_a, V_b, V_c and I_n/I_p, I_o/I_p, I_p/I_{load}, 1 hidden layer with 20 neurons and 11 outputs.The logsigmoid functions were used as the activation functions. The ANN was successfully trained to detect fault in the protective zone at 90 degree fault inception angle. While for testing the same inputs were used with 45 degree inception angle.

5) Results

Plotting of the confusion matrices is a means of testing the performance of the neural network for various types of errors that occurr during training and testing of the neural network. Fig. 4 plots the confusion matrices for the three phases of training, testing and validation and the “all confusion matrix” at the end is the resultant of the three individual confusion matrices. Correct classification by the neural network is indicated by the green diagonal cells in each matrix and the wrong classification by the red offdiagonal cells. In each of the matrices, the last cell in blue indicates the total percentage of cases that have been classified correctly in green and the vice-verca in red. Without introducing any delay in the fault samples, the overall accuracy was arround 80 percent. As the delay was intruced and increased in steps, the accuracy improved to 100 percent. The confusion matrices shown in Fig. 4 are for a delay of 25 samples, that is 12.5 ms. However, the accuracy on testing at a delay of 25 samples was only 94%.

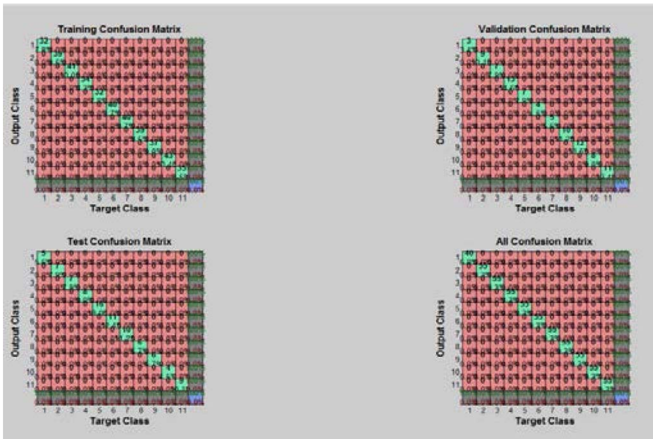

Fig. 4. Confusion matrices for Training, Testing and Validation Phases of ANN- 1 (25 sample delay)

B. ANN-2

1) Inputs

Line voltages and sequence current ratios were used as inputs to ANN-2. The six inputs used include V_{ab}, V_{bc}, V_{ca} and I_n/I_p, I_o/I_p, I_p/I_{load}. The outputs were eleven which consists of AG, BG, CG, AB, BC, CA, ABG, BCG, CAG, ABCG and no-fault. One hidden layer was used with different number of neurons for optimization.

2) Results

The numbers of iterations required for the training process were 48. The mean square error in fault detection achieved by the end of the training process was 4.57e-07 and the number of validation check fails was zero by the end of the training process.

Fig. 5 shows the training performance plot of the neural network.

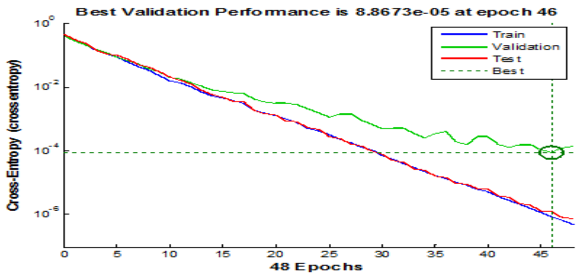

Fig. 5. Mean-square error performance of ANN-2 (6-20-11)

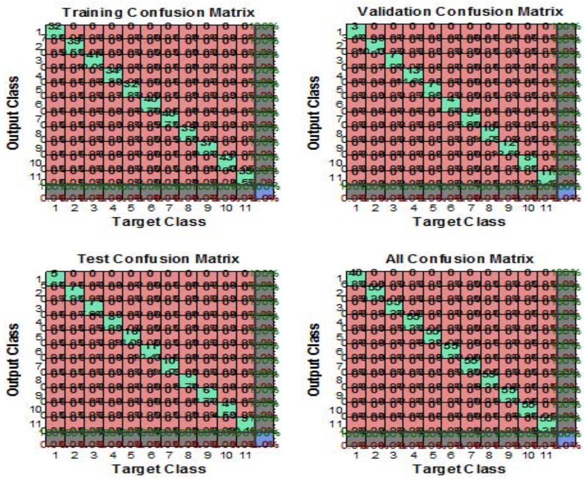

Fig. 6. Confusion matrices for Training, Testing and Validation Phases of ANN-2 (25 sample delay)

The accuracy of training for ANN-2 for fault classification with no delay was 79.8%. In order to increase the accuracy 5 samples were removed and the accuracy was still 83.2%. Then the accuracy reached a value of 94.1% at a delay of 10 samples, 99.4% at a delay of 15 samples, 99.2% at a delay of 20 samples. The accuracy of 100% was obtained at a delay of 25 samples as shown in Fig. 6. Further delay would make no sense. Confusion matrices for a delay of 25 samples (12.5 ms) are shown in Fig. 6.

Similarly ANN-2 was tested and the accuracy of testing for ANN-2 for fault classification with no delay was 81.3%. In order to increase the accuracy 5 samples were removed and the accuracy was still 84.9%. Then the accuracy reached a value of 93.5% at a delay of 10 samples, 98.0% at a delay of 15 samples, 98.8% at a delay of 20 samples. An accuracy of 100% was obtained at a delay of 25 samples as shown in Fig. 7.

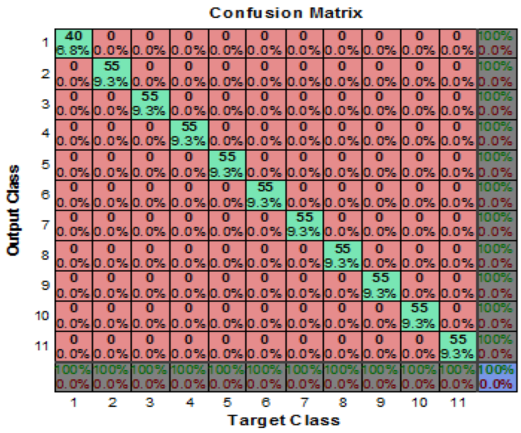

Fig. 7. ANN-2 Confusion matrix for testing of ANN-2 with 25 sample delay

C. ANN-3

1) Inputs

Phase voltages and sequence current ratios were used as inputs to ANN-3. The six inputs used include V_a, V_b, V_c and I_n/I_p, I_o/I_p, I_p/I_{load} for ANN-3.The outputs were eleven which consists of AG, BG, CG, AB, BC, CA, ABG, BCG, CAG, ABCG and no-fault. One hidden layer was used with different number of neurons for optimization.

2) Results

The numbers of iterations required for the training process were 52. The mean square error in fault detection achieved by the end of the training process was 3.47e-07 and the number of validation check fails was zero by the end of the training process.

Fig. 8 shows the training performance plot of the neural network with 6-20-11 configuration (6 neurons in the input layer, one hidden layers with 20 neurons and eleven neuron in the output layer)

The accuracy of training for ANN-3 for fault classification with no delay was 82.4%. In order to increase the accuracy 5 samples were removed and the accuracy was still 87.5%. Then the accuracy reached a value of 93.9% at a delay of 10 samples, 99.7% at a delay of 15 samples, 99.7% at a delay of

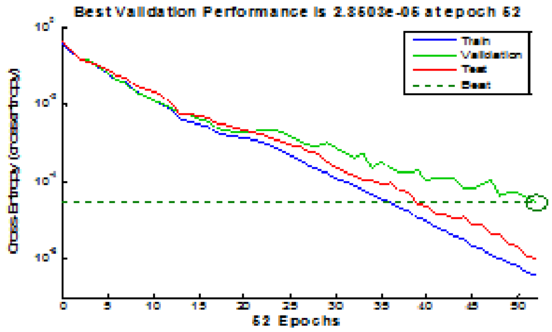

Fig. 8. Mean-square error performance of ANN-3 configuration

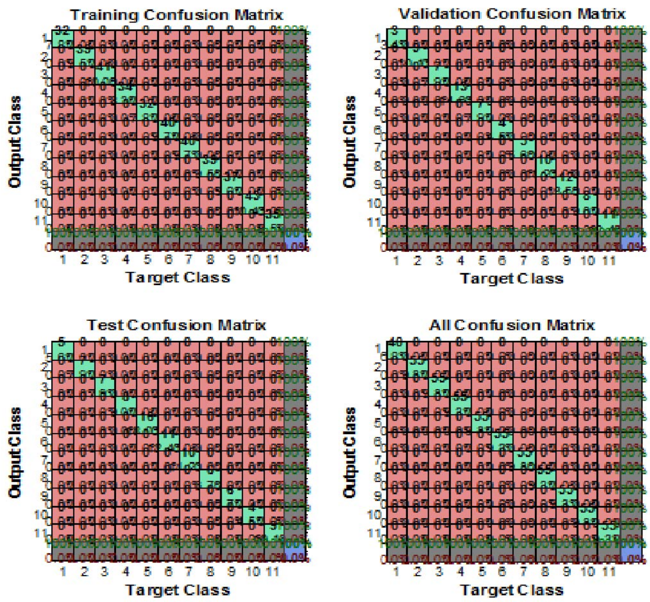

Fig. 9. Confusion matrices for Training, Testing and Validation Phase of ANN-3 (25 sample delay)

20 samples. An accuracy of 100% was obtained at a delay of 25 samples, as shown in Fig. 9.

Similarly ANN-3 was tested and the accuracy of testing for ANN-3 for fault classification with no delay was 83.0%. In order to increase the accuracy 5 samples were removed and the accuracy was still 79.1%. Then the accuracy reached a value of 85.5% at a delay of 10 samples, 86.7% at a delay of 15 samples, 89.4% at a delay of 20 samples. An accuracy of 90.5% (not 100%) was obtained at a delay of 25 samples, as shown in Fig.10.

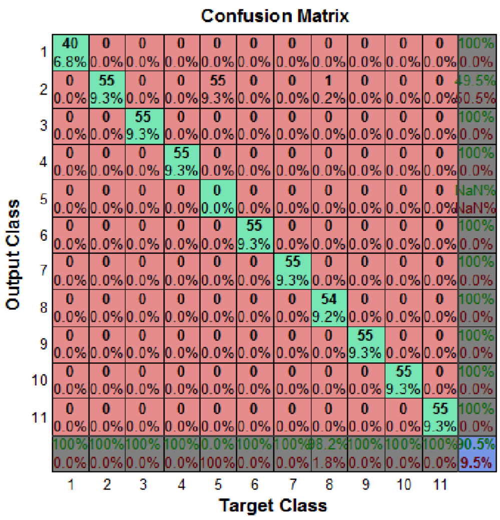

Fig. 10. ANN-3 Confusion matrix for testing of ANN-3 with 25 sample delay

IV. Comparison of Three ANN Configurations

In this part ANN-1, ANN-2 and ANN-3 are compared on the basis of the accuracy of fault classification on testing with variation in delay. The accuracy versus delay curves for the training and testing phases of ANN-1, ANN-2 and ANN-3 are given in figures 11, 12 and 13 ,respectively. It can be noted that only ANN-2 attains an accuracy of 100% on testing at a delay of 25 samples (12.5 ms) and is, therefore, the best ANN configuration.

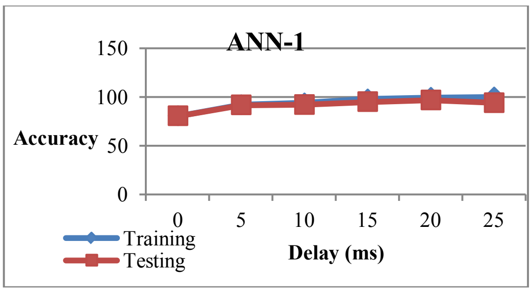

Fig. 11. Accuracy versus Delay for ANN-1

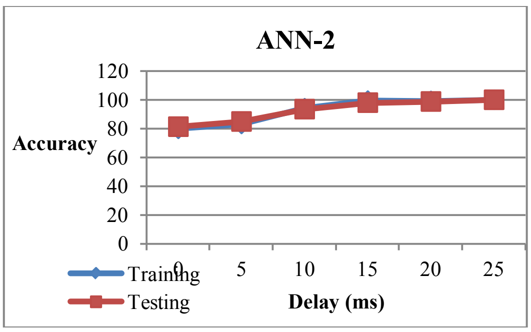

Fig. 12. Accuracy versus Delay for ANN-2

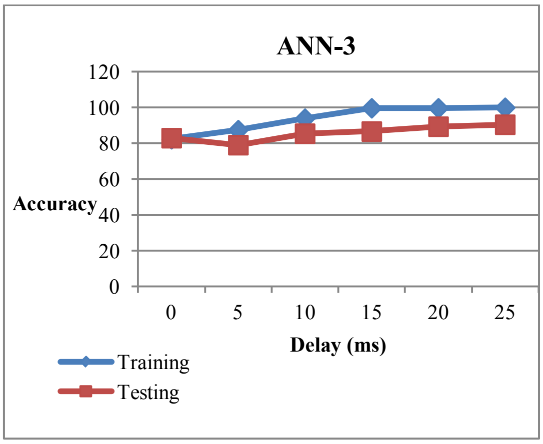

Fig. 13. Accuracy versus Delay for ANN-3

V. Training And Testing of ANN-2 With Full Data

ANN-2, being the best configuration among the three attempted configurations, was then evaluated with full data using two-fold cross verification technique. To that end, the complete data was divided into two equal data-sets, d_1 and d_2. The data set d_1 comprised the samples obtained for the fault inception angles of 0 degree and 90 degrees while d_2 comprised the samples obtained for 45 degree and 135 degree inception angles. First, d_1 was used for the training and then d_2 for the testing.

Fig. 14 shows the training performance plots of ANN-2. It can be seen that the network achieved the desired Mean Square Error (MSE) goal by the end of the training process. On testing with data set d_2, the accuracy of fault classification obtained without delay was only 77%, as shown in table I. The table and Fig. 15 also show the trend of accuracy improvement with increase in delay. An accuracy of 100% was achieved at a minimum delay of 25 samples, or 12.5 ms.

For cross verification of the performance of ANN-2, it was now trained with data set d_2 and then tested with data set d_1. This time too, an accuracy of 100% was obtained on testing with a minimum delay of 25 samples.

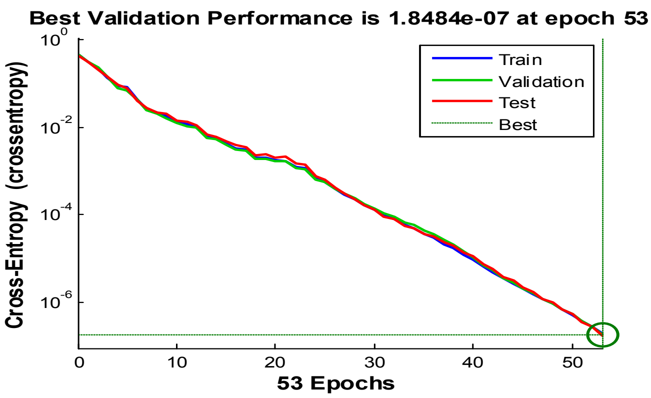

Fig. 14. Mean-square error performance of network (6-22-11)

TABLE I. ACCURACY VERSUS DELAY OF ANN-2 CONFIGURATION ON TESTING WITH DATA SET D2 AFTER TRAINING WITH DATA SET D1

S/No.	Delay (number of samples)	Accuracy (percent)
1	*0*	*77.0*
2	*10*	*88.2*
3	*15*	*96.1*
4	*20*	*98.2*
5	*25*	*100*

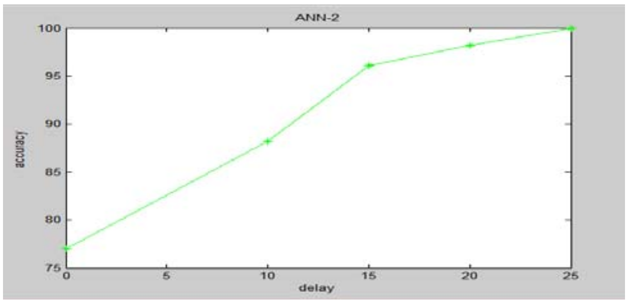

Fig. 15. Accuracy versus delay graph of ANN-2 on testing with d_2 after training with d_1

CONCLUSION

Detection and classification of faults in transmission lines using neural network has been attempted. Various kinds of faults namely phase-to-earth, two-phase-to-earth, phase-to-phase, three phase faults have been simulated in PSCAD/EMTDC. The data was generated for all the types of faults at three different locations (at 0 km, 50 km and 100 km of the line) with two values of fault resistance and eight values of fault inception angles at a sampling frequency of 2 kHz. The inputs to the neural networks are the rms phase voltages or line voltages and line currents or ratios of their sequence components. Back-propagation algorithm was used to train and test the networks. Three configurations of ANN have been attempted with different combination of inputs.

Relative performance of the three ANN configurations was checked using truncated data. ANN-2 among the three configurations has been found to be the best one with the minimum number of epochs and an accuracy of 100% for fault detection and classification for both training and testing with relay operating time of 12.5 ms (25 sample delay). Performance of ANN-2 was then evaluated with full data using the technique of two-fold cross verification. The results obtained were same as obatained with the truncated data.

An ANN based algorithm for the detection and classification of all types of faults on transmission line has been successfully developed which gives a better performance than that achieved by earlier researchers.

The work done can be extended for the compensated lines, that is, series-capacitor compensated lines and lines compensated with FACTS devices.

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